

**a** Sign-correct rate (bar > 50 %; blue above, red below)

|                      |             |             |             |             |             |               |             |               |
|----------------------|-------------|-------------|-------------|-------------|-------------|---------------|-------------|---------------|
| $\alpha = 0.25$ (§7) | 38 %<br>3/8 | 50 %<br>3/6 | 60 %<br>3/5 | 75 %<br>3/4 | 40 %<br>2/5 | 50 %<br>2/4   | 40 %<br>2/5 | 47 %<br>7/15  |
| $\alpha = 0.5$       | 75 %<br>6/8 | 67 %<br>4/6 | 40 %<br>2/5 | 25 %<br>1/4 | 40 %<br>2/5 | 25 %<br>1/4   | 20 %<br>1/5 | 73 %<br>11/15 |
| $\alpha = 1.0$       | 62 %<br>5/8 | 67 %<br>4/6 | 20 %<br>1/5 | 0 %<br>0/4  | 40 %<br>2/5 | 25 %<br>1/4   | 40 %<br>2/5 | 47 %<br>7/15  |
|                      | iid         | yaw -37.5°  | yaw -22.5°  | yaw +22.5°  | yaw +37.5°  | planar 4.5 cm | cam -5°     | cam +5°       |

**b** Median off-target ratio (bar  $\leq 0.25$ ; min 1.62, all 24 cells fail)

|                      |      |            |            |            |            |               |         |         |
|----------------------|------|------------|------------|------------|------------|---------------|---------|---------|
| $\alpha = 0.25$ (§7) | 4.17 | 2.64       | 5.14       | 21.41      | 2.52       | 3.98          | 3.53    | 4.96    |
| $\alpha = 0.5$       | 4.38 | 2.52       | 2.58       | 3.97       | 7.99       | 2.90          | 2.29    | 3.53    |
| $\alpha = 1.0$       | 6.08 | 3.10       | 1.62       | 2.52       | 3.65       | 4.43          | 7.57    | 5.03    |
|                      | iid  | yaw -37.5° | yaw -22.5° | yaw +22.5° | yaw +37.5° | planar 4.5 cm | cam -5° | cam +5° |

condition cell; entries show the rate and k/n valid pairs. The  $\alpha = 0.25$  row is derived from the §7 per-pair evidence, the  $\alpha = 0.5$  and 1.0 rows from the frozen 9b grid